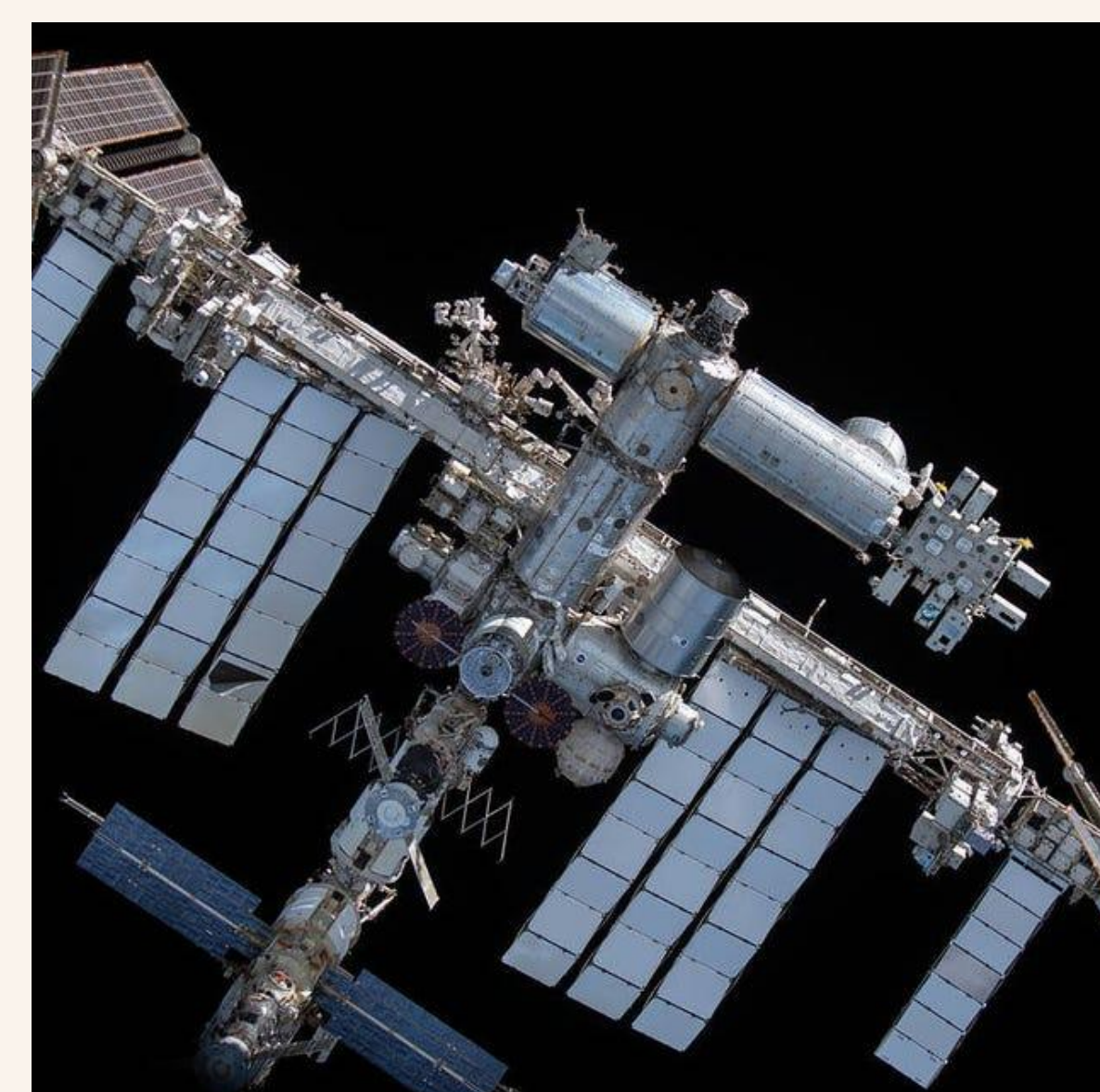
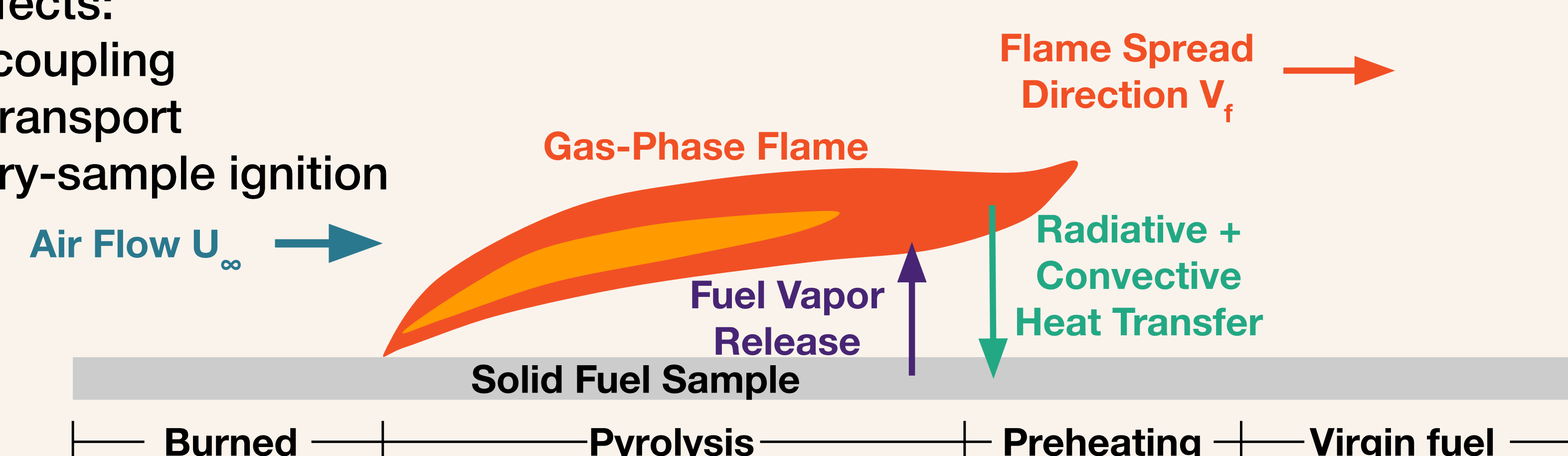


INTRODUCTION

- Flame spread occurs when **heat** causes **nearby solid fuel to pyrolyze** and **release combustible gases**.
- In microgravity, reduced buoyancy makes flame behavior more dependent on **forced airflow** and confinement.
- In spacecraft, materials are closely packed, and limited fire response options make fire propagation especially hazardous.
- This study uses computational modeling to examine how **separation distance H** between parallel fuel samples affects:
 - thermal coupling
 - oxygen transport
 - secondary-sample ignition



ISS, Image courtesy of NASA



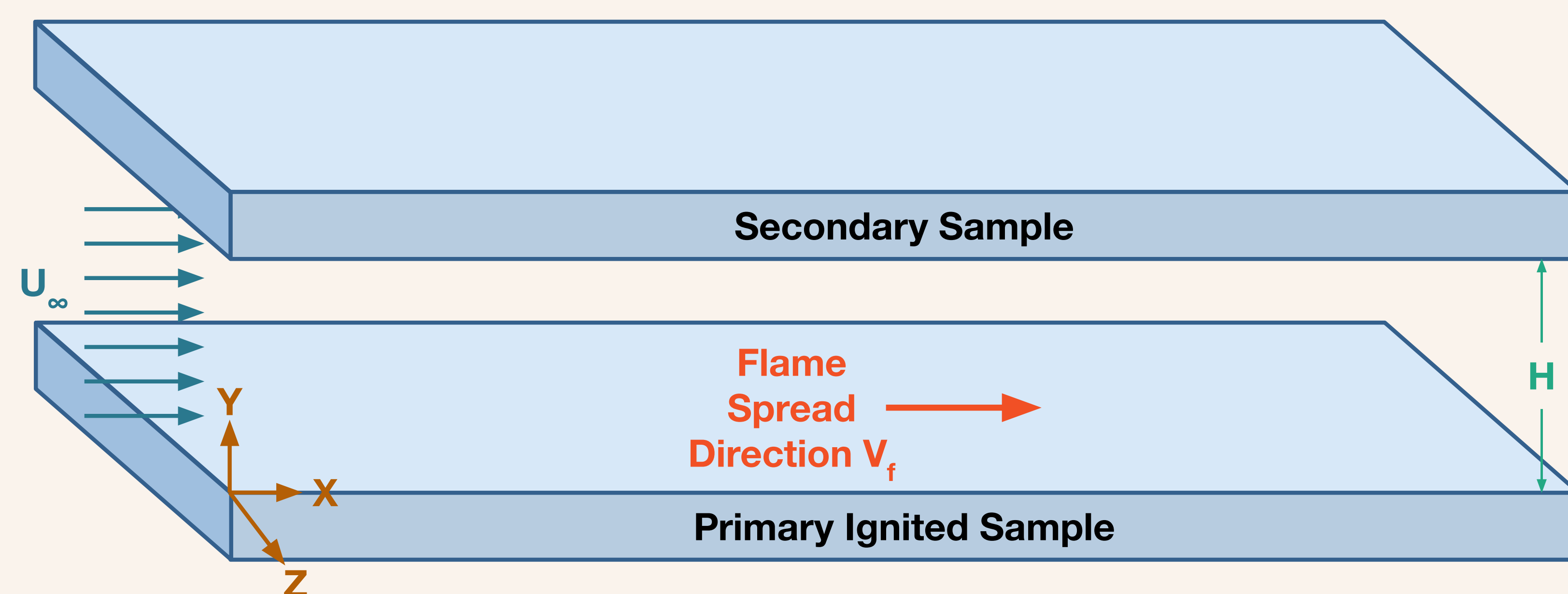
COMPUTATIONAL SETUP

Numerical Model

- Transient conservation equations** solved for mass, momentum, energy, and species.
- Gas-phase combustion** represented using finite-rate, second-order Arrhenius.
- SN3D solves the **radiative transfer equation** for gases and solid surfaces, providing radiative heat feedback.
- SIMPLER algorithm couples pressure and velocity.
- Solid model solves **heat conduction** and **two-step pyrolysis**, producing fuel vapor and surface mass loss.
- Gas and solid solutions are advanced together using adaptive time steps.

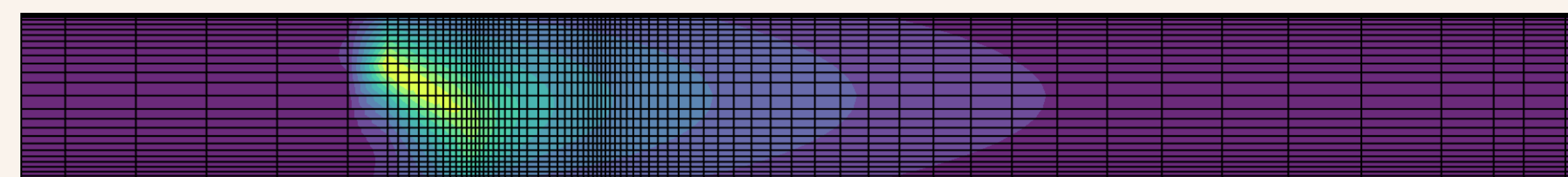
Simulation Conditions

- Fuel: **SIBAL composite** (75% cotton and 25% fiberglass)
- Separation distances: H=1.0, 1.5, 2.0, and 2.5cm**
- Forced airflow:** $U_\infty=6\text{cm/s}$, $g=0$
- Only the primary sample is externally ignited, the secondary sample heats through flame feedback.



Adaptive Mesh

- Nonuniform mesh with finer cells near the samples and flame region.
- Adaptive regriding follows the flame as it spreads downstream.

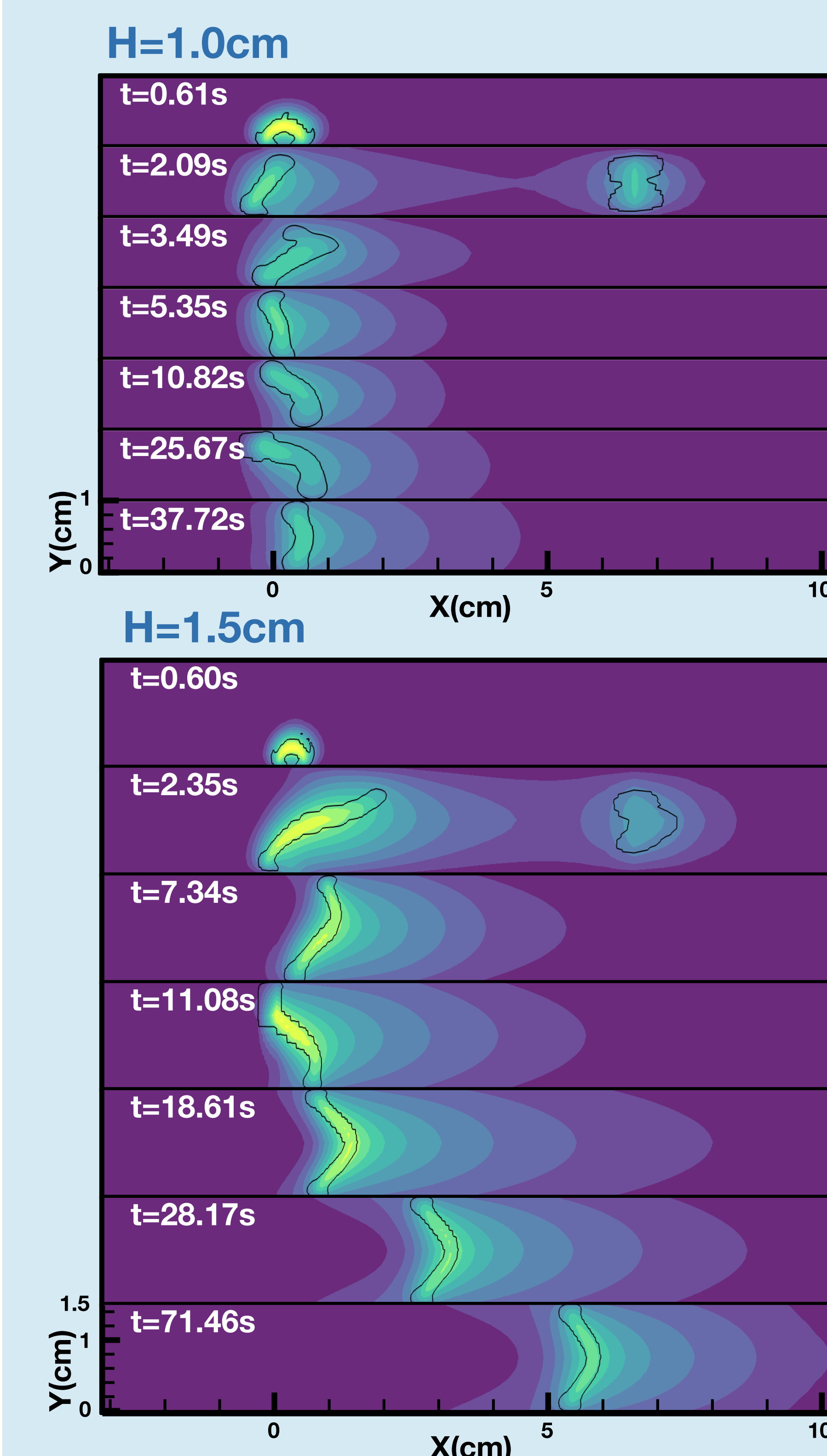


ANALYSIS

Temperature Evolution

Temperature contours were analyzed as time progresses.

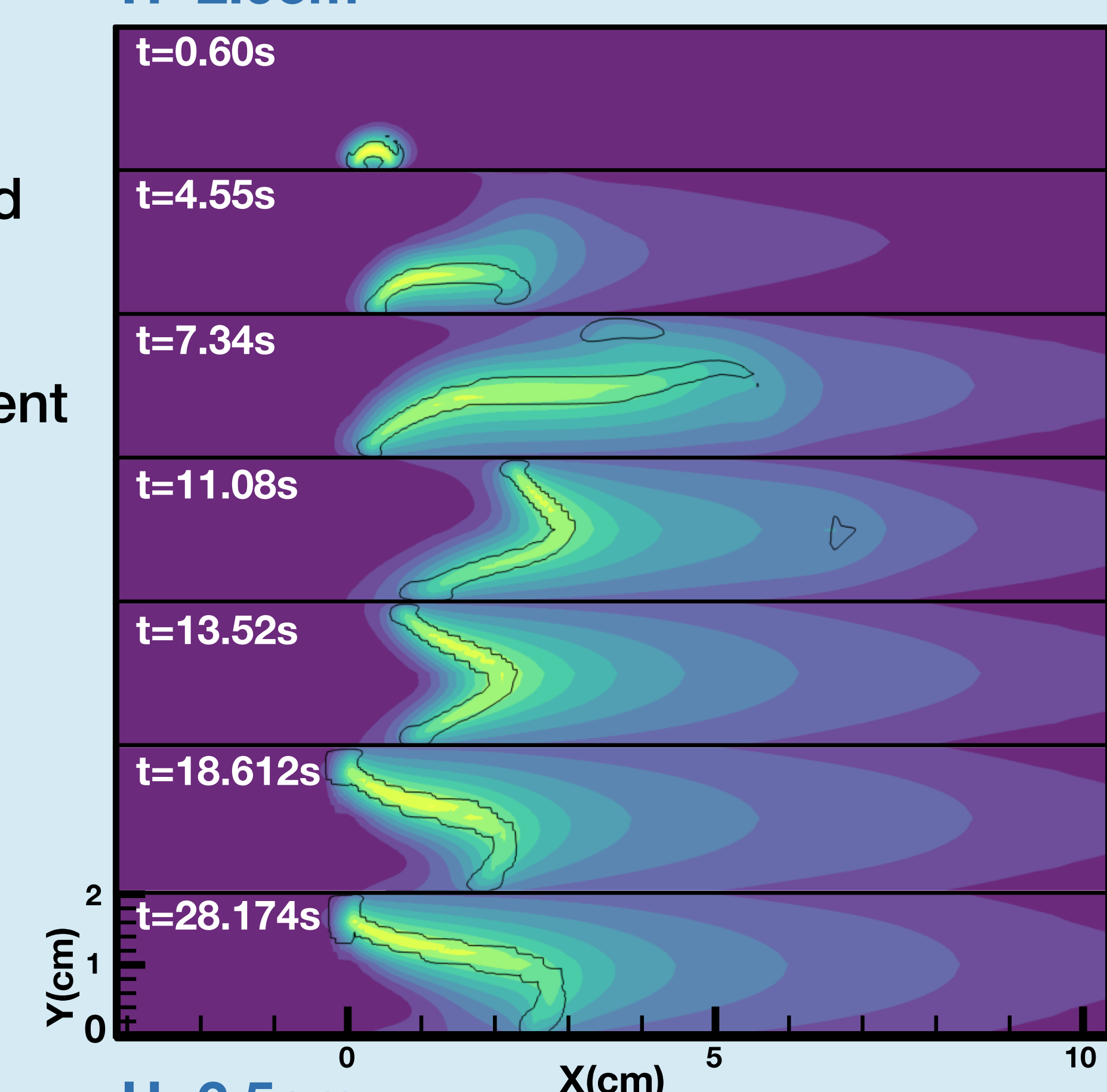
- Smaller gaps** had curved and confined flame structures, **stronger interaction**.
- At H=2.5cm, the flame remained elongated and attached to the primary sample.
- H=1cm flame became more compact and cooler, consistent with stronger confinement and oxygen depletion.



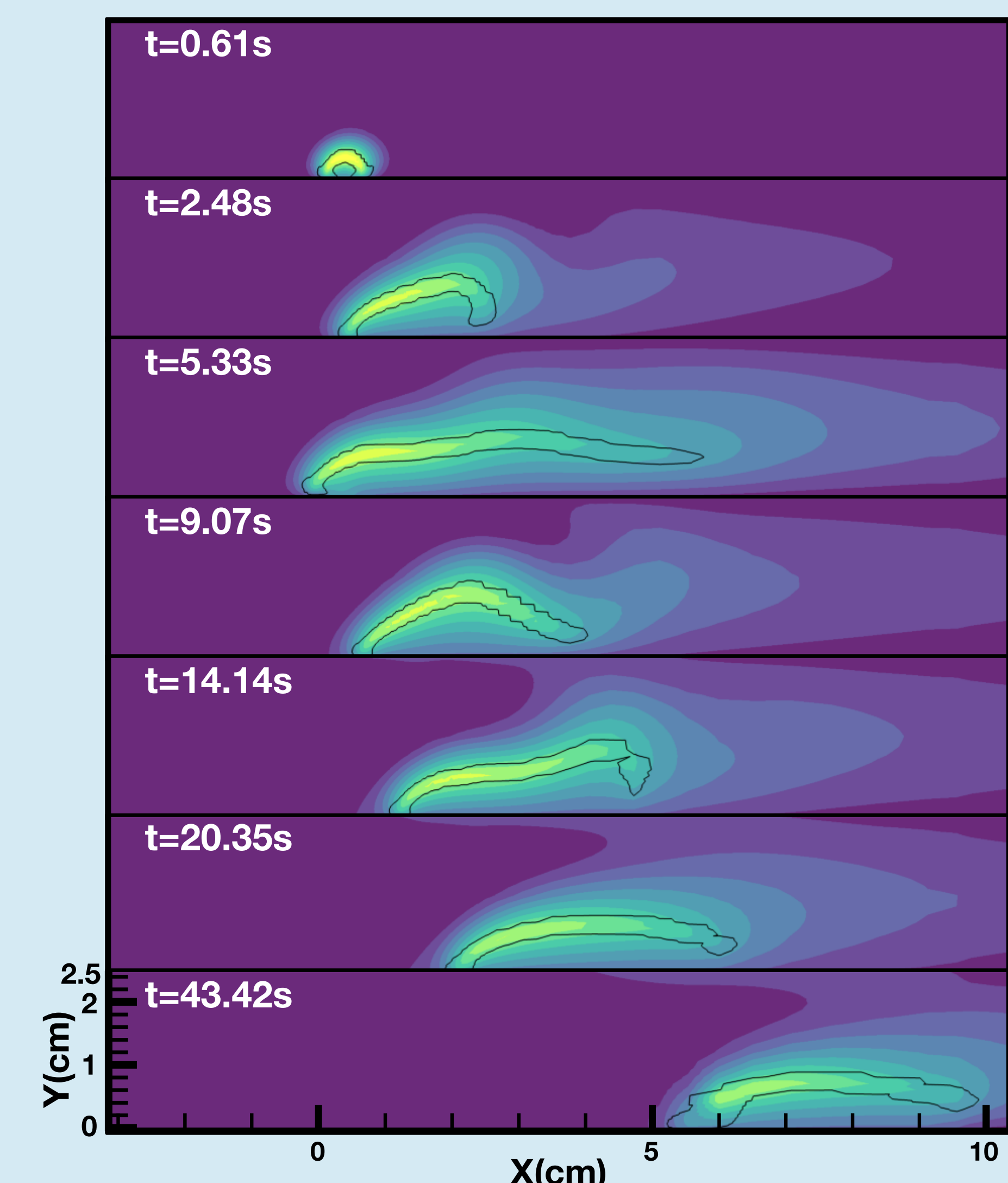
Black line: $\omega'=10^{-4}\text{gcm}^{-3}\text{s}^{-1}$ reaction rate contour.

T (°C) 200 400 600 800 1000 1200 1400 1600

H=2.0cm

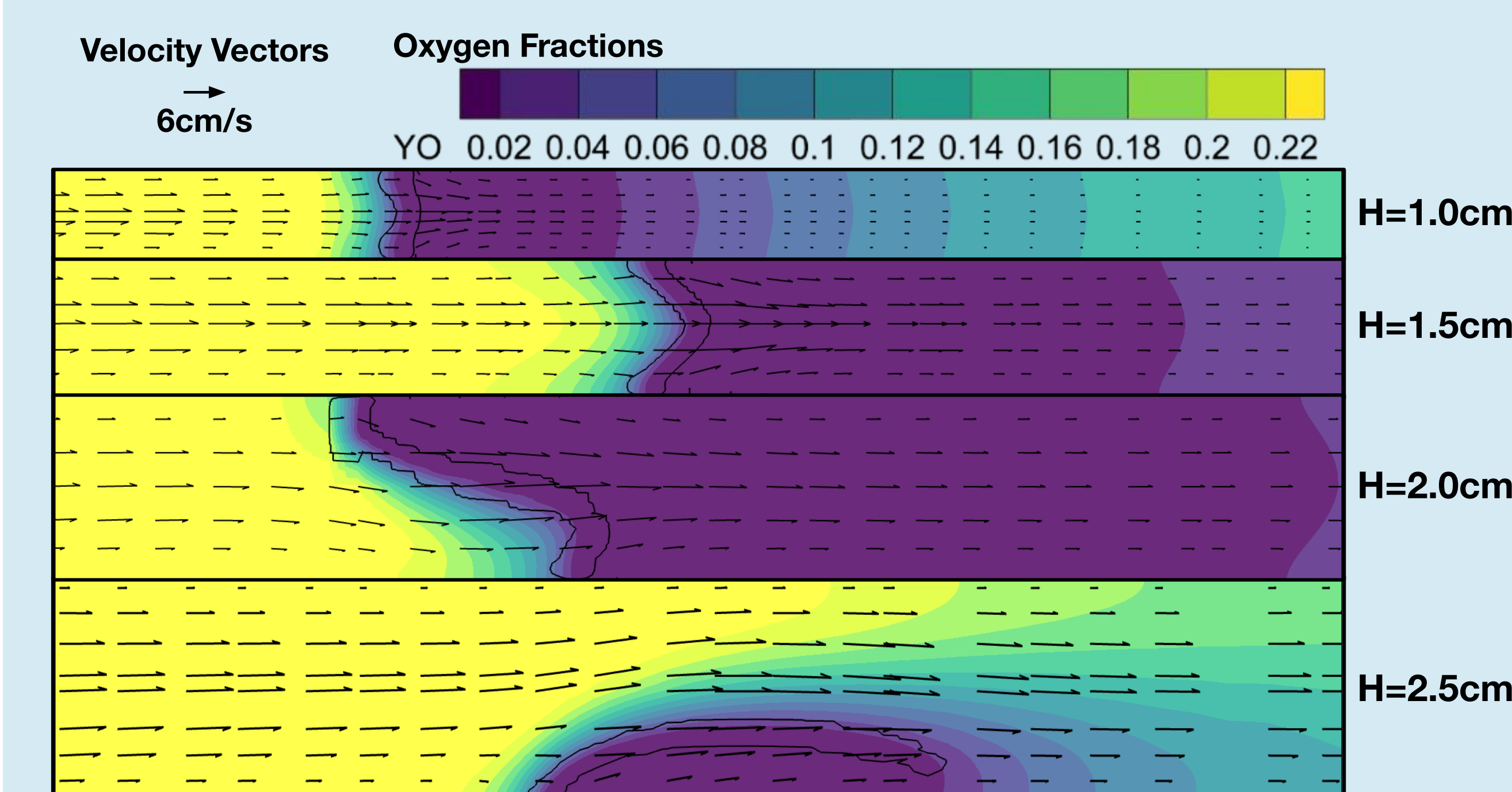


H=2.5cm



Oxygen Transport and Flow

Snapshots were taken when flames reached a **semi-steady state**.

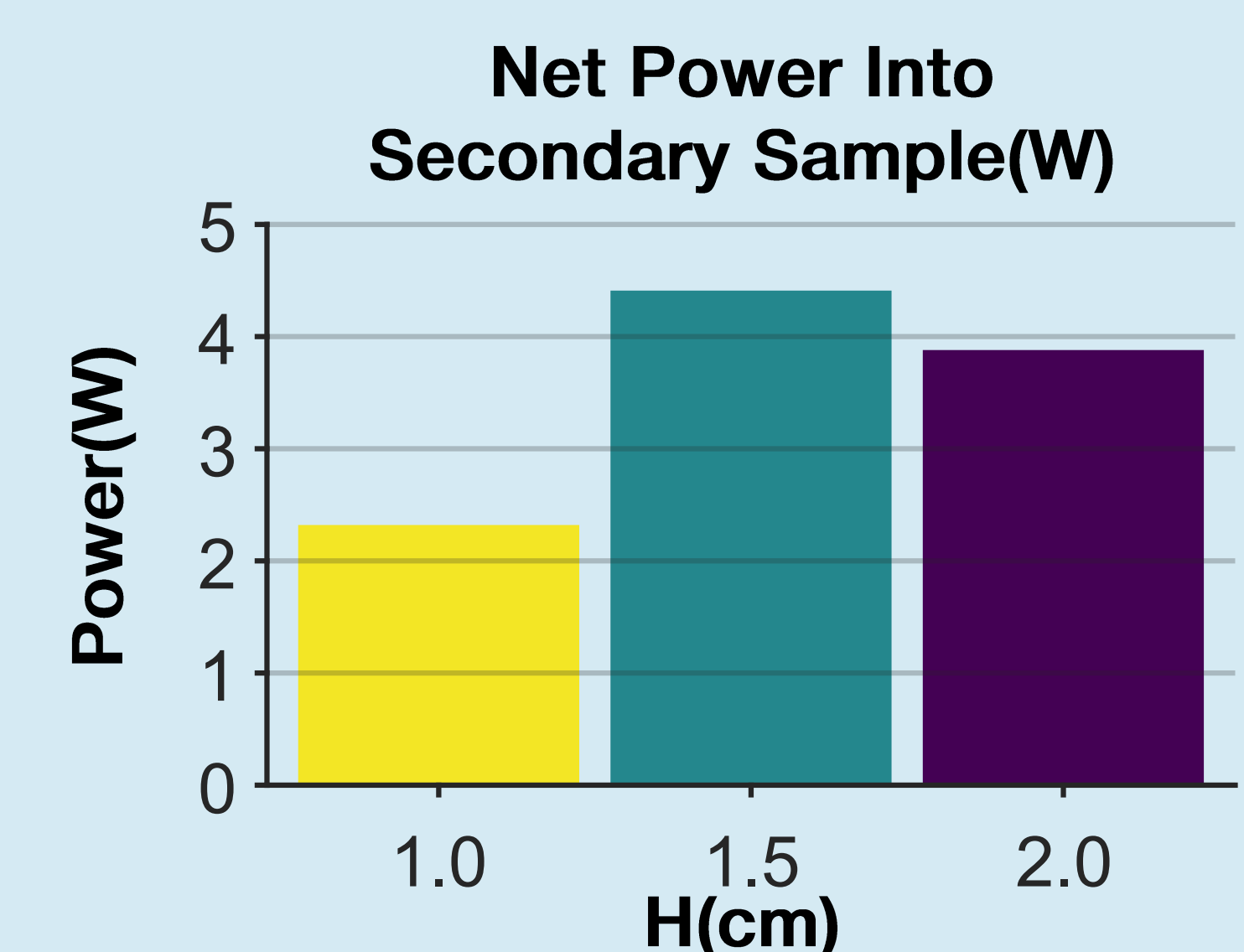


- At smaller separations, the flame occupied a larger fraction of the gap, **limiting oxygen transport** toward the secondary sample.
- Despite greater oxygen availability, the H=2.5cm secondary sample did not ignite → **heat transfer is the limiting factor**.
- The flame redirects the imposed flow and develops along regions where fuel vapor and oxygen overlap.

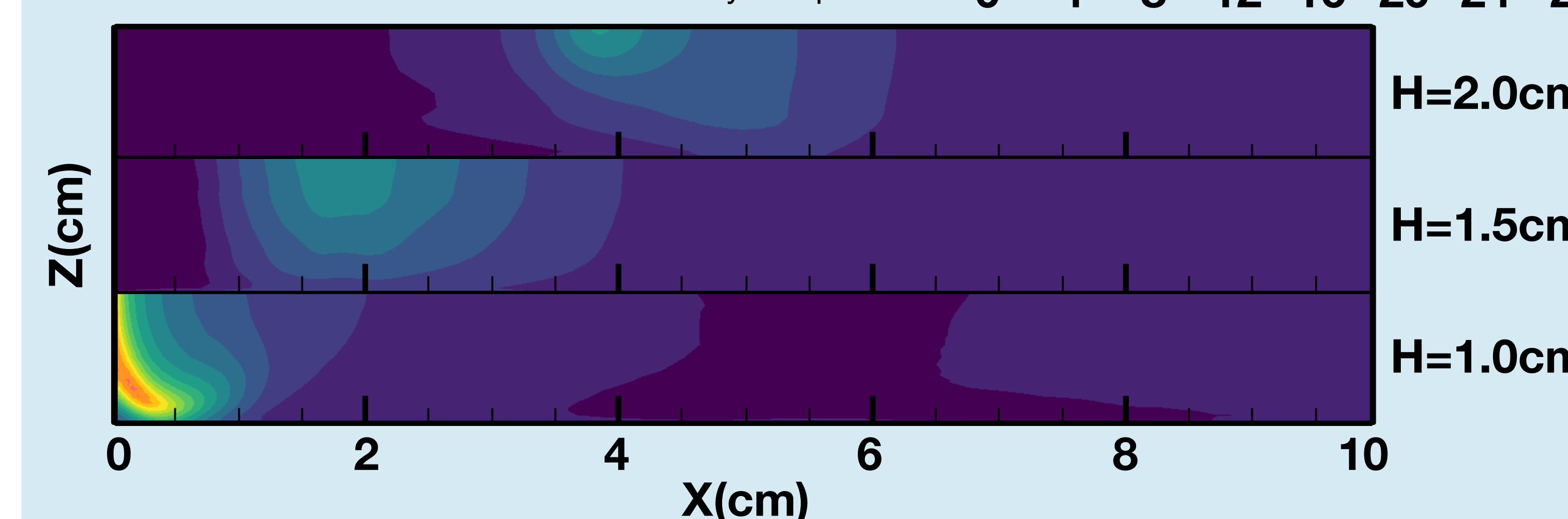
Pre-Ignition Heat Transfer

- H=1.5 had the greatest net heat transfer rate into the secondary sample.
- Although H=1.0cm produced the highest local heat flux, the heating was **highly localized**, resulting in the lowest net power.
- The broader heating distribution at H=1.5cm produced **greater total thermal feedback**.
- The **non-monotonic result** indicates that decreasing separation does not necessarily increase overall heat transfer to the secondary sample.

H(cm)	Ignition Time(s)	Average Net Flux(kW/m²)
1.0	6.599	1.06
1.5	5.829	2.00
2.0	13.517	1.76



Net Heat Flux Into Secondary Sample (kW/m²) Over the modeled half-width of the secondary sample



CONCLUSION

- Secondary ignition occurred for H=1.0, 1.5, and 2.0cm, but was not observed at H=2.5cm.
- The earliest secondary ignition occurred at H=1.5cm, indicating the **most favorable tested balance** between heat feedback and oxygen availability, which creates **greater heat transfer hazard**.
- Intermediate separation produced the greatest overall pre-ignition heat transfer, while the smallest separation produced a more intense but highly localized hot spot.
- The H=2.5cm case remained well ventilated but was too weakly thermally coupled to ignite the secondary sample.
- The results indicate three cases: confined/oxygen-depleted, strongly coupled, and well-ventilated but heat-transfer limited.**

NEXT STEPS

- Extend the study to multiple imposed airflow speeds while keeping the same separation distances.
- Investigate theoretical correlation between flame spread rate and the effects of H and U_∞ .

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